

Day 37 - Classification Algorithms in Machine Learning

1) Logistic Regression - Detailed Explanation

Logistic Regression is a supervised machine learning classification algorithm used to predict categorical outputs.

It is mainly used for binary classification problems, where output has only two classes.

Examples:

Spam / Not Spam

Pass / Fail

Yes / No

Disease / No Disease

Although the name contains "regression", it is actually used for classification.

Why is it called Logistic Regression?

Because it uses a logistic (sigmoid) function to convert output into a probability between 0 and 1.

Output Meaning:

If probability is near 1 $\rightarrow$ class = 1

If probability is near 0 $\rightarrow$ class = 0

2) Mathematical Implementation of Logistic Regression

Logistic Regression first calculates a linear equation:

$$z = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

Then it applies the Sigmoid Function:

$$\hat{y} = 1 / (1 + e^{(-z)})$$

Where:

$\hat{y}$ = predicted probability

e = Euler's number

z = linear combination of inputs

Example of Logistic Regression

Suppose a student's study hours decide whether they pass or fail.

Formula:

$$z = -4 + 1.5x$$

If student studies 4 hours:

$$z = -4 + 1.5(4)$$

$$z = -4 + 6 = 2$$

Now apply sigmoid:

$$\hat{y} = 1 / (1 + e^{(-2)})$$

$$\hat{y} \approx 0.88$$

Result:

Probability of passing = 0.88 (88%)

Since it is above 0.5, prediction = Pass

Advantages of Logistic Regression

- Simple and easy to understand

- Works well for binary classification

- Fast to train

- Gives probability output

Disadvantages

- Not suitable for very complex data

- Works best when relationship is somewhat linear

3) K-Nearest Neighbors (KNN)

KNN stands for K-Nearest Neighbors.

It is a supervised machine learning algorithm

used for both:

- classification

- Regression

But it is mostly used for classification.

How KNN Works

KNN works by finding the nearest K data points to a new data point.

Then it checks which class is most common among those neighbors.

Example:

Suppose we want to classify a fruit as:

Apple

Orange

The algorithm checks the nearest fruits based on:

weight

color

size

If most nearby fruits are apples, then
prediction = Apple

Steps in KNN

choose value of K

calculate distance between points

Find nearest neighbors

check majority class

Predict output

Distance Formula (Euclidean Distance)

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This is used to find how close one point is to another.

Advantages of KNN

- Simple and easy to understand

- No training phase required

- Works well on small datasets

Disadvantages

- Slow for large datasets

- Sensitive to noise

- Requires feature scaling

4) Support Vector Machine (SVM)

SVM stands for Support Vector Machine.

It is a supervised machine learning algorithm used for:

- Classification

- Regression

But it is mostly used for classification.

How SVM Works

SVM tries to find the best boundary (hyperplane) that separates different classes.

Example:

Suppose we want to classify:

Cats

Dogs

SVM draws a line or boundary that best separates the two groups.

Main Idea

SVM chooses the boundary with the maximum margin, meaning the gap between classes should be as wide as possible.

Important Terms:

Hyperplane → decision boundary

Support Vectors → nearest data points to the boundary

Margin → distance between support vectors and boundary

Advantages of SVM

Works well for high-dimensional data

Effective for classification

Good for small to medium datasets

Disadvantages

Hard to understand mathematically

Slower on large datasets

Needs proper tuning

5) Naive Bayes

Naive Bayes is a supervised classification algorithm based on Bayes Theorem.

It is called "Naive" because it assumes that all features are independent of each other.

Bayes Theorem Formula

$$P(A|B) = (P(B|A) \cdot P(A)) / (P(B))$$

Where:

$P(A|B)$ = probability of A given B

$P(B|A)$ = probability of B given A

$P(A)$ = prior probability

$P(B)$ = evidence

How Naive Bayes Works

It calculates the probability of each class and predicts the class with the highest probability.

Example:

Email classification:

Spam

Not Spam

If an email contains words like:

"Free"

"offer"

"Win"

Then probability of Spam becomes high.

Types of Naive Bayes

Gaussian Naive Bayes - for continuous data

Multinomial Naive Bayes - for text data

Bernoulli Naive Bayes - for binary features

Advantages of Naive Bayes

Very fast

Easy to implement

Works well for text classification

Good for spam filtering

Disadvantages

Assumes features are independent

Less accurate if assumption is wrong

6) Decision Tree

Decision Tree is a supervised machine learning algorithm used for:

Classification

Regression

It works like a tree structure with questions and decisions.

How Decision Tree Works

It splits the data step by step based on conditions.

Example:

Predict whether a student will pass or fail:

Study hours > 5 ?

Yes $\rightarrow$ Pass

No $\rightarrow$ Fail

This creates a tree-like decision process.

Parts of Decision Tree

Root Node $\rightarrow$ first question

Decision Node $\rightarrow$ condition-based split

Leaf Node $\rightarrow$ final output

How Splitting Happens

Decision Tree chooses the best split using:

- Gini Index

- Entropy

- Information Gain

These help find the best feature for splitting.

Advantages of Decision Tree

- Easy to understand

- Easy to visualize

- Works for both numerical and categorical data

Disadvantages

- Can overfit easily

- Unstable for small changes in data

7) Random Forest

Random Forest is an ensemble learning algorithm made up of many decision trees.

It is used for:

- Classification

- Regression

How Random Forest Works

Instead of using one decision tree, Random

Forest creates multiple decision trees and combines their outputs.

For classification:

It uses majority voting

For Regression:

It uses average prediction

Example

Suppose 5 decision trees predict:

Tree 1 $\rightarrow$ Pass

Tree 2 $\rightarrow$ Pass

Tree 3 $\rightarrow$ Fail

Tree 4 $\rightarrow$ Pass

Tree 5 $\rightarrow$ Pass

Final output:

Pass (because majority says Pass)

Why Random Forest is Better than Decision Tree

Because:

It reduces overfitting

It gives better accuracy

It is more stable

Advantages of Random Forest

- High accuracy

- Reduces overfitting

- Works well on many datasets

- Handles missing values better than some algorithms

Disadvantages

- More complex

- Slower than a single decision tree

- Harder to interpret